Light Mate: Smart Chessboard

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GitHub (includes Demo Video and Final Report)

1. INTRODUCTION

Traditional chess offers a rich physical experience, while digital chess provides powerful analysis and accessibility. However, players must often choose between the two. This project presents ***Light Mate***, a smart chessboard that combines a physical board with embedded electronics to enable real-time move detection, visual feedback, and online gameplay integration.

2. OBJECTIVES

- Detect chess piece movement in real time using Hall-effect sensors.
- Provide visual feedback using addressable RGB LEDs.
- Validate moves using a Raspberry Pi and Python logic.
- Enable online gameplay through Lichess API integration.
- Deliver a seamless physical + digital chess experience.

3. SYSTEM OVERVIEW

System pipeline: Sensors → I²C expanders → Raspberry Pi → SPI LED control → Visual feedback.

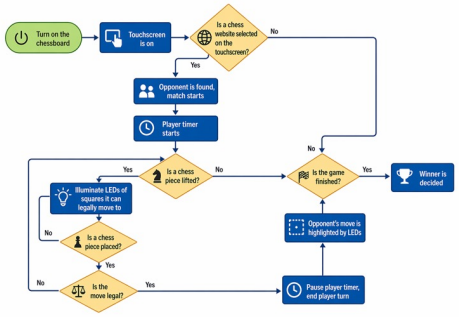

Figure 1: Software Flow Diagram

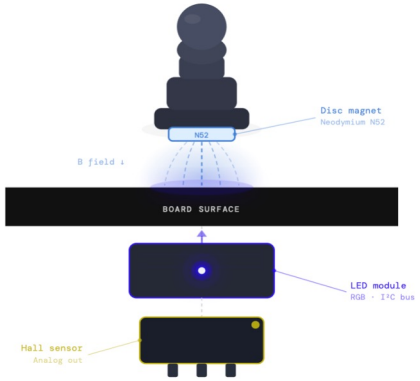

Figure 2: System Architecture Overview

4. HARDWARE DESIGN

- **64 Hall-effect sensors (DRV5033):** Detects presence of chess pieces using magnets embedded underneath each piece. Each sensor outputs a digital signal indicating whether a piece is on a square, forming the core input system for board state detection.
- **81 SK9822 addressable RGB LEDs:** Provide real-time visual feedback, such as highlighting legal moves, opponent moves, and board states.
- **4 MCP23017 I/O expanders:** Extend the Raspberry Pi's limited GPIO pins using I²C communication. They allow all 64 Hall sensors to be read efficiently, acting as the interface between sensors and the processor.
- **SN74HCT125 level shifter:** Converts 3.3V logic signals from the Raspberry Pi to 5V for the LEDs. Ensures safe and reliable communication between the Pi and LED chain without damaging components
- **Touchscreen Display (Waveshare 3.5"):** Provides a graphical user interface via HDMI and touch input. Used for game setup, debugging, and user interaction.
- **Raspberry Pi 4:** The main processing unit of the system. Runs Python code to read sensors, validate moves, control LEDs, and handle game logic + connectivity.
- **On-board push button:** For user input (system reset, mode selection, and interaction control).
- **Custom 2-layer PCB:** integrates all components into a single compact board. Handles signal routing, power distribution, and physical layout, ensuring reliability and scalability of the full 8×8 system.

5. SOFTWARE DESIGN

The system operates using a real-time loop implemented in Python.

- Continuously reads sensor states via I²C.
- Detects piece lift and placement events.
- Uses python-chess to validate legal moves.
- Updates LEDs to highlight valid destinations.
- Generates opponent moves for demonstration.

PIPELINE: Read Sensors → Detect Move → Validate → Update LEDs

Software implemented in Python on Raspberry Pi 4

6. RESULTS

Figure 3: CAD Render of Expanded Enclosure

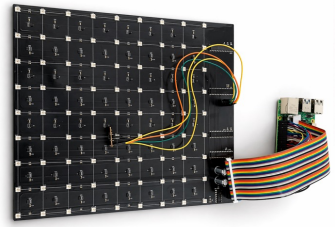

Figure 4: Soldered PCB connected to Pi 4 and button module

Figure 5: Final Product

7. FUTURE WORK

- **Miniaturized Compute (CM4 Integration):** Replace Raspberry Pi 4 with a Compute Module integrated into a custom PCB to reduce wiring and enable a smaller, more compact enclosure.
- **Stronger Chess Engine:** Integrate advanced engines (e.g., Stockfish) for real-time analysis, move evaluation, and training feedback.
- **Wireless & App Connectivity:** Add mobile/web interface for game tracking, remote play, and cloud synchronization.
- **Portable Design:** Implement battery power and optimized power management for a fully portable system.
- **Screen:** Increase size and implement touch interface.
- **Enclosure:** Use better materials and improve durability. There were compromises taken due to time and budget constraints. The project required larger 3D printers that were not available and printing through a third party was not an option due to budget limits.

Figure 6: CM4 (Left) VS Pi 4 (Right)

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